



Islamic University Of Technology

EEE 4308 : Digital Electronics

Project Name : SAP – 1

Team : Digital Electronauts

Phase - E

Team Advisor: Mr. Hayder Jahan Parash

Team Members:

Name	Student ID
Ar- Rafi Ishraq	210021330
Ashfiq UL Rahman	210021342
Shahria Salim Forhad	210021315
Tahsik Islam	210021318
Shafayat-ul-firoz	210021312
Md. Rezwan Islam Sabbir	210021305
Bayzid Mohammod Sakib	210021250

Objective:

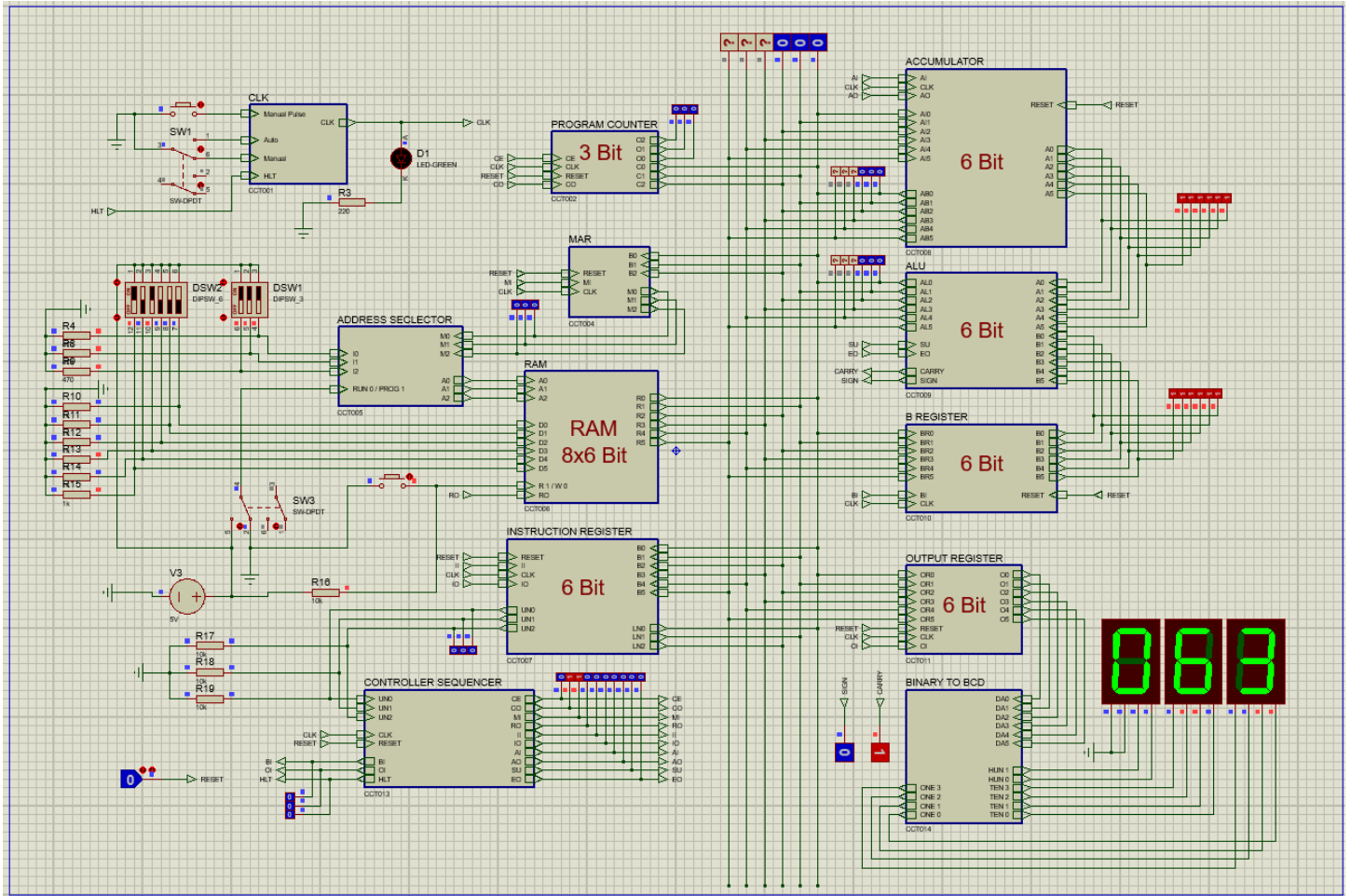
Complete integration of SAP-1

Description:

This project is about understanding of 6 bit computer and construction of it. Basically its about understanding inside process of a computer. We can understand what does it mean clock speed of a computer and how we can control this speed. For running this computer we need to program the RAM first, then we can run the computer. Program is stored inside RAM as a set of instruction. Other modules acts according to instruction. Program counter counts the instruction and MAR sends the memory address for next instruction. Instructed operation is occurred inside ALU. Value of ALU is stored inside Accumulator. When necessary , Accumulator sends the data to output register to show the result. Necessary signal for all the module is sent by controller sequencer. Controller sequencer generates signals according to instruction code sent by instruction register. All the module is runed by clock pulse. Without clock pulse this computer can't work. There is HLT operation in our computer that stopes the main clock, as a result whole computer stopes. This is total working system of our SAP-1.

Phase - E

Complete SAP in Proteous:



Discussion:

It is the complete design of our SAP project. We integrated all the module with BUS. At first it wasn't running. We had to go through a lot of trouble shooting. We found some small mistake. Main problem was with our controller sequencer. So, we had to debug our sequencer. After completing the debug phase, we ran our SAP. All the problem were solved. Now our complete SAP is running perfectly.

Contribution of each member:

ID: 210021330:

In this phases I designed ALU, Controller sequencer and binary to BCD converter & Digital display. Among all those Controller sequencer was the most hardest one for me. Controller sequencer is a sequential circuit. I had to do lot of calculations to find out the equations. After finding those equations I built the circuit, as usual the circuit wasn't working so I had to do debugging. Then it worked. ALU design was the interesting one. I added a feature so that it is able to subtract a big number from a small number. For designing the converter and display I went through some YouTube tutorials and websites. I used Double Dabble method to made the converter, and used 7 segment as our digital display.

ID: 210021342:

MAR or memory address register is a register responsible for storing memory address of the data. To complete the construction of MAR, I had to use three control pins like three CONN-SIL-3, one TBLOCK-12 & one 74HC173. At first, I faced some issues regarding the use of 74HC173. Then I went through some youtube videos & found how to do this. In phase C,D,E I made input unit, Address selector and helped in making the RAM.

ID: 210021315:

"CLOCK" was one of the most important part among phase A & B. For designing the CLOCK I used Two 555 Timer ic, Two 7400 IC (NAND gate) .First I had to design a Astable 555 timer circuit as astable pulse signal then I had to design Monostable 555 timer as selection pin and lastly I had to design Bistable 555 timer as manual pulse signal. Then they were connected by two nand gate as such that the two signals can be controlled by selection pin. In phase C,D,E I helped in making the ALU.

ID: 210021318:

The Output Register in the SAP-1 is a part of the Output Unit. It is a special register used to temporarily hold the data that needs to be output to external devices, such as a display or other peripherals. The Output Register is controlled by the control unit of the SAP-1, which manages data transfer between the memory, CPU, and output devices.

When the SAP-1 computer needs to send data to an output device, the data is transferred from the CPU's internal registers or memory into the Output Register. From there, it can be sent to the external device under the control of the control unit.

We used two 74HC173 (4 bit register) and a bunch of resistors to build this circuit. We also used some LEDs to demonstrate the outputs.

ID: 210021312:

The accumulator was not that easy to make at first, after understanding how it is working it started to make sense and then it became doable. I have used two 74173 ICs and a 74HC245 IC. I have also used two 7404 ICs (Not gate). Because of using the ICs the PCB design became quite simpler and easy to make. The circuit is running perfectly and there are no errors showing in both the schematic and PCB part.

ID: 210021305:

The program counter which is a part of the control unit counts from 000 to 111, each combination indicates the binary address the program that is stored in that runs, as for 6-bit sap, I used 74HC181 IC and 74HC245 IC to design program counter. The 3 diode LEDs were used to indicate which memory address it is accessing. The pin diagrams were a bit difficult as I messed up 1st time trying it to do fast but after fixing the connection the program counter worked as it should.

ID: 210021250:

B – Register was the simplest one among all the modules. For making B-Register I used Two 74HC173 IC and one NOT Gate (7404) IC. I faced difficulties in finding proper resources for making this circuit, finally I found one. After fixing some small issues circuit is working correctly.